COMPLETED QUOTIENTS OF HENSELIZATIONS COME FROM FINITE TYPE ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The easy half of the algebraization toolkit: it produces the finite type model whose formal completion is a prescribed completed quotient of a henselization.

Editorial note. The label lemma-algebraize-easy names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- More on Algebra, Lemma more-algebra-lemma-henselization-Noetherian-pair: the henselization of a finite type algebra over a Noetherian ring with respect to an ideal is Noetherian
- More on Algebra, Lemma more-algebra-lemma-henselization (construction): the henselization is a filtered colimit of etale C-algebras C-prime with C/IC isomorphic to C-prime/IC-prime
- Standard henselization fact recorded by the author: such etale extensions have the same henselization and the same completion as the pair (C, IC)

Source reference. Original lemma 0A02: sources/stacks-chow/tags.txt, source lines 12620–12620. Prerequisite pointer invoked at 1465; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AGV: sources/stacks-chow/tags.txt, source lines 13208–13208. Prerequisite pointer invoked at 1461; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let I be an ideal of a Noetherian ring A. Let C^h be the henselization of a finite type A-algebra C with respect to the ideal IC. Let $J \subset C^h$ be an ideal. Then there exists a finite type A-algebra B such that $B^\wedge \cong (C^h/J)^\wedge$.*

Proof. By More on Algebra, Lemma 0AGV the ring C^h is Noetherian. Say $J = (g_1, \ldots, g_m)$. The ring C^h is a filtered colimit of étale C algebras C' such that $C/IC \to C'/IC'$ is an isomorphism (see proof of More on Algebra, Lemma 0A02). Pick an C' such that $g_1, \ldots, g_m$ are the images of $g'_1, \ldots, g'_m \in C'$. Setting $B = C'/(g'_1, \ldots, g'_m)$ we get a finite type A-algebra. Of course (C, IC) and $C', IC')$ have the same henselizations and the same completions. It follows easily from this that $B^\wedge = (C^h/J)^\wedge$. □

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